On the effectiveness of stop-loss rules. An analytical framework based on modeling overnight gaps and the stationary bootstrap

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Abstract

A stop-loss rule is a risk management tool whereby the investor predefines some condition that, upon being triggered by market dynamics, imply the liquidation of her outstanding position. Such a tool is widely used by practitioners in financial markets with the hope of improving their investment performance by cutting losses and consolidating gains. But, do stop-loss rules really add value to an investment strategy? In this work we give an answer to this question for four popular implementations of stop-loss rules using two different statistical methods, which complement each other and lead to a more robust answer. On the one hand, we use a model-based approach in which we present two different models for the price of a New York Stock Exchange equity, which consider the phenomenon of overnight gaps; on the other hand, we complement the previous approach with a data-based framework by using the stationary bootstrap to obtain different replicas of the financial time series corresponding to the historical price of several NYSE stocks.

As a general conclusion we find that, in rising markets, stop-loss rules improve the expected risk-adjusted return according to most metrics, while improving absolute expected return in falling markets. Furthermore, we find that in general the fixed percentage stop-loss rule may be the most powerful among the popular rules that we consider in this work, followed by the RSI-based stop-loss rule.

Keywords: Stop-loss; Risk Management; Financial modeling; overnight gap; bootstrap.

1 Introduction

A *stop-loss* order is an order an investor may place so that her position is liquidated the moment a certain pre-specified condition (set according to a *stop-loss rule*) is met by market dynamics. Whether the position the investor holds is short or long, the purpose of setting a stop-loss is to cut losses and consolidate gains, being this a most basic yet popular tool for risk management.

The importance of placing stop-loss orders cannot be overestimated. The fall of Banco Popular in Spain (June 2017) is a very recent example of the dangers of not

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incorporating stop-loss orders into one's investing strategy: investors that bought-and-hold hoping for an eventual trend reversal lost 100% of their investment, as Banco Santander bought Banco Popular's four billion shares for EUR 1, driving the individual share price to zero. Additionally it has been argued that the setting of stop-loss orders guards investors from the disposition effect [Silva and Da Silva, 2015], which refers to the observed tendency of investors to hold on to losing positions much longer than they hold to winning positions.

Several researchers have looked at the problem of discerning the effectiveness of stop-loss orders in various contexts. We mention a few that are of particular interest to our work. The paper by Acar and Toffel [2001] studies how a stop-loss rule affects the returns distribution, by assuming the asset follows a Brownian Random Walk with drift, and then evaluating the financial profitability of a simple stop-loss strategy under the previous assumption. Kaminski and Lo [2014] follow a similar approach: they developed a rigorous analytical framework for measuring the impact of simple 0/1 stop-loss-re-entry rules on the expected return and volatility of an arbitrary portfolio strategy (again assuming that assets follow a Random Walk) and provide an empirical analysis of a stop-loss rule applied to a buy-and-hold strategy in U.S. equities. James and Yang [2010] on the other hand, do not assume that financial assets obey a particular model, but instead base their analysis on the use of the stationary bootstrap as a tool to replicate financial time series adequately.

None of the previously mentioned works, and others that we surveyed related to the assessment of the value of stop-loss rules, considers the oftentimes observed large variation in the price across non trading hours. This clearly affects the correct triggering of a stop-loss rule due to the by-passing of the established stopping-time.

The early works of French [1980], Gibbons and Hess [1981] and Keim and Stambaugh [1984], showed that returns on non trading days present different distribution than returns on trading days. In particular, they showed that in the U.S. market the expected returns of stocks are significantly lower or even negative on Mondays compare to returns on preceding Fridays, a phenomenon that has been coined the *weekend effect*.

Cooper et al. [2008] found that in the more recent price history in U.S. markets overnight returns (including weekends) are consistently positive while daytime returns are close to zero or negative; thus, concluding that the U.S. equity premium on the first decade of the 21st century is solely due to overnight returns. Wiener and Tompkins [2008] extended the study of return differences between trading and non-trading hours to higher moments of the returns distribution and for European and Japanese markets, and found that in general the distribution of non-trading periods (or overnight) returns displayed a higher degree of non-normality compared to that of the trading periods returns. This lead these researchers to conclude that while trading periods returns may follow some diffusion process, the non-trading periods returns follow a jump process. This fact was also argued by Geman et al. [2001] who studied asset prices arising from market clearing conditions.

The present work contributes to the analysis of the value of stop-loss rules in various novel aspects, considering previous studies on the subject and on the different distribution displayed by non-trading periods returns as opposed to trading periods returns. First, and most important, we implement models for the behavior of asset prices that take into account the existence of overnight gaps in prices, that is jumps

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from the Closing price of one day to the Open price of next trading day. Second, as a complement to our asset model-based analysis, we consider a non-parametric approach based on the stationary bootstrap, thus providing an alternative point of view for assessing the advantages of stop-loss rules. Additionally we contribute with two new performance measures, the Return-VaR and the Return-ES ratios, based on the VaR and Expected Shortfall risk measures. And last but not least for all our experiments we use higher-frequency financial data (hourly prices). Our ultimate goal is to provide solid evidence for the conjecture that good stop-loss strategies may provide higher risk-adjusted returns, with respect to a passive Buy-and-Hold strategy.

The rest of this paper is structured as follows: Section 2 presents the general framework for the analysis of stop-loss rules, and the four specific stop-loss rules that we deal with in our simulations. Section 2.1 contains our two new performance metrics, RVaR and RES ratios, plus three other popular measures to assess the impact of applying to a simulated trading period each of the four stop-loss policies considered in our study. Section 3 presents two financial price models somewhat more sophisticated than the traditional random walk or ARMA models, but without including yet overnight gaps. The intention of this intermediate step, between the traditional models and our proposed models that include overnight gaps, is to give a comparative analysis of the impact on the performance measures of including some quantification of overnight gaps in the modeling of stock's time series. We make the comparisons by running the same trading simulations on the models with and without consideration of overnight gaps. Section 4 presents the two financial price models that consider overnight gaps. Section 5 briefly describes the stationary bootstrap resampling method. Sections 6 and 7 contain our experiments and out-of-sample results based on our models for simulating return behavior, and on the bootstrap, both using high-frequency financial data. Section 8 concludes.

2 Stop-loss rule and stop-loss policy

In this section we set up the general framework for the analysis of stop-loss rules. We will focus on long-only strategies, and will not take into consideration slippage (difference between order price and execution price) or transaction costs. As a matter of fact, although it is true that in practice there might be a significant negative correlation between equity performance and its bid-ask spread, in our study this is not an issue because we have deliberately selected the 30 most transacted stocks in 2013, which makes them some of the most liquid too, and so it is safe to assume that the bid-ask spread at the time when the stop-loss is triggered will be similar to the bid-ask spread that the investor would encounter at the end of the investment horizon when the position is liquidated under a buy-and-hold passive strategy; as for transaction costs these are irrelevant to our study, because the transaction costs associated with the stop-loss strategies that we consider are exactly the same as those associated with a buy-and-hold strategy, for two reasons: first, because we only consider pure stop-loss policies, i.e. we disregard re-entry rules; second, because placing a stop-loss order is free, regardless of which broker the investor operates with.

Our formal definition of a stop-loss policy is a simplified version of the stop-loss-re-entry trading scheme of Kaminski and Lo [2014] because we are interested in analyzing how different rules for exiting the market can better prevent us from losing money

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than just staying in for a period of time irrespective of market movements. However, our definition allows for different criteria for stopping losses, as opposed to Kaminski and Lo [2014] where the only criterion considered by the authors is that cumulative returns reach a threshold, a form of the *fixed percentage barrier* criterion as considered below.

Given an asset A, we consider a *stop-loss rule* as a decision criterion SL_t on the price history of A such that at every time t it takes the value 0 if the criterion is not satisfied or 1 otherwise. We denote by $\nu(SL_t)$ the value (0 or 1) of SL_t. We can formalize SL_t as a boolean formula on the real numbers, arithmetic symbols, order and other relations as needed. However, we prefer to skip this formality and show instead what is meant with an example. Consider the criterion of cumulative returns of a certain asset reaching a given barrier. Let R_t be the cumulative returns at time t, and β the value of the barrier at any time. Then the stop-loss rule in this case is described by the boolean formula:

$$SL_t := R_t \geq \beta$$

This expression takes value $\nu(SL_t) = 1$ if SL_t holds (i.e. at time t, R_t has reached or surpassed the barrier β), or value $\nu(SL_t) = 0$ if SL_t does not hold.

Then, given a time horizon T and an initial time instant $t_o < T$ at which we buy a risky asset A, a *stop-loss policy* is a risk management measure that depending on the value of a stop-loss rule it signals to exit the investment at a certain time t, $t_o \leq t \leq T$, where then we shall sell all of our holdings in A and use the proceeds to go long on a risk-free asset F up until time T. We formalize this idea as follows.

Definition 2.1 *Given a stop-loss rule SL_t, a* ***stop-loss policy*** $\mathbf{s}(\vec{x_t}, \vec{\gamma})$ *for a risky asset A with returns $\{r_t\}$ is a binary wealth-allocation scheme $\{s_t\}$ between A and a risk-free asset F with return $\{r_f\}$, in such a way that either we are fully invested in A ($s_t = 1$) or totally withdrawn from A ($s_t = 0$) and invested in F. Formally, at initial time t_o, $s_{t_o} = 1$, and for $t : t_o < t \leq T$,*

$$s_t = s(\vec{x_t}, \vec{\gamma}) = \begin{cases} 1, & \text{if } \nu(SL_t) = 0 \text{ and } s_{t-1} = 1 \text{ (stay in)} \\ 0, & \text{if } \nu(SL_t) = 1 \text{ and } s_{t-1} = 1 \text{ (exit)} \\ 0, & \text{if } s_{t-1} = 0 \text{ (stay out)} \end{cases} \tag{1}$$

where $\vec{x_t}$ is a vector of known information about A up until time t, and $\vec{\gamma}$ is a vector of parameters that guide the stop-loss rule SL_t.

In our experiments the risk-free asset F is a AAA government bond, for example, a 1-year U.S. government bond. As in [Kaminski and Lo, 2014], in our parametric models we assume that the returns $\{r_t\}$ of the risky asset A under consideration satisfy the following property:

> The expected return μ of A is greater than the risk-free rate r_f, and let $\pi := \mu - r_f > 0$ denote the risk premium of A.

This property simply excludes the perverse case where the stop-loss policy adds value just because the risk-free asset that the investor transfers the capital to has a

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higher expected return than A. In our parametric models we ensure this property holds by setting the abovementioned rates to appropriate values. The risk-free rate is set to the historical mean return of a 1-year U.S. government bond, which, during the period 1990-2017 has been 3.171%. On the other hand, the expected return of our hypothetical asset is set to the average return of the NYSE during the same period (1990-2017), which is 6.14%. Assuming compounding, this implies an hourly return of $3.942924 * 10^{-5}$.

In our study we implement and analyze the performance of the following four stop-loss rules (SL_t):

Fixed percentage barrier (%): $SL_t := P_t < P_{t-1}^{max}(1-a)$, where P_{t-1}^{max} is the highest price achieved until time $t-1$ and the positive quantity $a \cdot 100\%$ is the maximum percentage of that price P_{t-1}^{max} that the investor is willing to lose in its position. Note that this corresponds to a mobile support barrier common among practitioners. The value of a can be adjusted from historical data or be given a constant value based on a certain risk profile. We tune a according to each of the five performance metrics to be considered (see Section 2.1).

Average True Range (ATR): $SL_t := P_t < P_{t-1} - \alpha ATR_d$, where usually $\alpha \in [1.5, 3]$, $d \leq t$ and ATR_d is a crude estimation of daily historical volatility in the recent past

$$ATR_d := \frac{1}{N}\sum_{i=0}^{N} TR_{d-i}$$

with $TR_d := max\{High_d - Low_d, High_d - Close_{d-1}, Close_{d-1} - Low_d\}$, $High_d$, Low_d and $Close_d$ are the maximum, minimum and closing price at time d, and a typical value for N is 14 days. Note that whilst ATR_d is measured on a daily basis, SL_t can be hourly or follow other higher time frequency.

Relative Strength Index (RSI): The RSI is a momentum indicator that compares the magnitude of an asset's recent gains and losses over a specified time period. It is defined as

$$RSI_t(w) := 100 - \frac{100}{1 + RS(w)_t} \in [0, 100]$$

where $RS(w)_t$ is the average gain of up periods during the specified time frame, divided by the average loss of down periods during the specified time frame; that is, for a window of w periods (usually 7 or 14 trading days), containing u up (U) and $w-u$ down (D) periods,

$$RS(w)_t := \frac{\frac{1}{u}\sum_{i \in U}(P_i - P_{i-1})}{\frac{1}{w-u}\sum_{i \in D}(P_i - P_{i-1})}$$

RSI values of 70 or above indicate that a security is becoming overbought or overvalued, and therefore may be primed for a trend reversal or corrective pullback in price. The rationale for this strategy may lie in the belief that too much euphoria can in fact anticipate a change of regime. This behavior is in fact what one observes in major market corrections [Sornette, 2004]. In this case the stop-loss rule is: $SL_t := RSI_t(w) \geq 70$.

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Triple Moving Average crossover (MA): A popular exit signal is the triple crossover of short, medium and long-term moving averages (MA). Practitioners use many different MA periods, and one that is common and seems reasonable based on our own experience would be the triplet 5-20-70. Assuming the starting scenario is that the shorter-period MAs are above the longer-period ones, the exit signal is given by the cumulative event of the 5-period MA crossing the 20 and the 70-period MAs, together with the 20-period MA crossing the 70-period MA. If those three events happen, there is a high probability of a continued fall in price. This method's rationale is that longer-term MAs show more of an asset's historical price trend, whereas shorter-term MAs show more the asset's recent price trend, and so if the recent trend crosses the historical trend, we may be witnessing a change of regime. A MA of p periods is defined as

$$MA(p) := \frac{1}{p}\sum_{i=0}^{p-1} C_{t-i}$$

where C_t is the asset's closing price at session t. The stop-loss rule in this case reads, for $a = 5$, $b = 20$ and $c = 70$,

$$SL_t := MA(a) < MA(b) \wedge MA(b) < MA(c)$$

2.1 Performance metrics

We will assess the impact of a stop-loss policy (based on each of the four stop-loss rules presented) on investment performance with five metrics. The first one is simply the expected return, and therefore there is no penalty for the risk the investment strategy involves. The remaining four metrics do penalize for higher risk. The first two are well-known risk-adjusted performance measures, and the last two are new proposals we are contributing:

Expected return: The expected return of a risky asset A with returns $\{r_{s_t}\}$ or $\{r_t\}$, depending whether a stop-loss policy is in place or not respectively, is given as usual by $\mathbb{E}[r_{s_t}]$, $\mathbb{E}[r_t]$ respectively. Note that,

$$\begin{aligned}\mathbb{E}[r_{s_t}] &= \mathbb{E}[r_{s_t}|s_t = 0] \cdot \mathbb{P}(s_t = 0) + \mathbb{E}[r_{s_t}|s_t = 1] \cdot \mathbb{P}(s_t = 1) \\ &= r_f \cdot \mathbb{P}(s_t = 0) + \mathbb{E}[r_{s_t}|s_t = 1] \cdot \mathbb{P}(s_t = 1)\end{aligned}$$

Because $\mathbf{s}$ is a dynamic binary wealth-allocation rule, the probabilities above are best understood as the fraction of time in which the investor is either stopped out or invested in the risky asset. When the former occurs, the investor switches to the risk-free asset until the end of the investment horizon.

Sharpe Ratio (SR): The Sharpe Ratio of a stop-loss policy [Sharpe, 1994], is the expected return of the risky asset in excess of the risk-free rate divided by the standard deviation of that risky return, for a given time horizon (usually one year):

$$SR := \frac{\mathbb{E}[r_t] - r_f}{sd(r_t)} \tag{2}$$

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The Sharpe ratio is a popular metric in finance to compare asset performance, but it is far from being ideal. Due to the fact that it uses the standard deviation as a measure of risk, this metric treats downside risk the same as upside risk, which in general is not desirable. The following three alternative measures address the issue of penalizing only for downside risk.

Sortino Ratio (SOR): The Sortino Ratio of a stop-loss policy [Sortino, 1994], is the expected return in excess of the minimum return rate acceptable (e.g. the risk-free rate) T divided by the so-called Downside Risk (DR), for a given time horizon (usually 1 year).

$$SOR := \frac{\mathbb{E}[r_t] - T}{DR} \tag{3}$$

The Downside Risk is a measure that is conceptually similar to the standard deviation of the returns (as a random variable), although it is not exactly the same because the reference value is not an expected return, but a target return, and its support is restricted to values below target. Mathematically,

$$DR := \sqrt{\int_{-\infty}^{T} (r_t - T)^2 f(r_t) dr_t}$$

where $f(r_t)$ is the density function of the returns in the buy-and-hold strategy.

Return-VaR ratio ($RVaR$): The $RVaR$ of a stop-loss policy is the median return (in excess of the risk-free rate) divided by the median return minus the 1-year 5% VaR:

$$RVaR := \frac{median(r_t) - r_f}{median(r_t) - VaR_{5\%}} \tag{4}$$

where $VaR_{5\%} = CDF_{r_t}^{-1}(5\%)$, CDF being the cumulative distribution function of the returns generated by the stop-loss strategy.

Return-ES ratio (RES): The RES of a stop-loss policy is the median return, in excess of the risk-free rate, divided by the median return minus the 1-year 5% Expected Shortfall:

$$RES := \frac{median(r_t) - r_f}{median(r_t) - ES_{5\%}} \tag{5}$$

where $ES_{5\%} := \frac{1}{5\%} \int_0^{5\%} VaR_\gamma \, d\gamma$, which effectively computes the average VaR_γ with γ ranging from 0 to 5%.

Naturally, under the stop-loss policy $\mathbf{s}$, r_{s_t} replaces r_t in the above definitions.

In both the two new performance metrics that we propose ($RVaR$ and RES), we choose the median return instead of the expected return for two reasons: first, because the median is a metric that is more robust to outliers than the mean; second, because a denominator involving percentiles (VaR or ES) clearly favors the use of the median instead of the mean; then, the fact that the ratio as a whole becomes easier to interpret if both the numerator and denominator have the same metric justifies the use of the median in the numerator too. The ratio can then be interpreted as follows: the

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numerator informs about how larger is the median return of the risky investment or strategy compared to the return of a risk-free asset, while the denominator takes some of those merits off by penalizing for downside risk.

One may argue that this numerator is harder to interpret, as the expected return is a concept that most investors are perfectly acquainted with and may be easier to interpret from a utility-theoretic perspective. However, by using the median return we provide a different perspective, which has a powerful interpretation: by using the median return, the numerator of our ratio is informing about the minimum amount by which the return of the risky investment surpasses that of the risk-free asset 50% of the time.

Similarly to the Sortino ratio, our *RVaR* and *RES* ratios exclusively penalize for downside risk, which is an important advantage over the Sharpe ratio.

Finally, our *RVaR* and *RES* ratios have at least two advantages over the Sortino ratio. On the one hand, there is less arbitrariness involved in the choice of the value of the parameter involved: in the case of the *RVaR* and *RES* ratios, the parameter involved is the percentile of the distribution of returns to be chosen, and reasonable values are essentially just three (5, 2.5, 1), which contrasts with the much higher discretion involved in the choice of the target return in the case of the Sortino ratio. On the other hand, the denominator of the *RVaR* and *RES* ratios is nonzero with probability one, which is something not guaranteed in the Sortino ratio, as this heavily depends on the choice of the target return, among other factors.

3 Models of asset prices without overnight gap

Traditional models for equity prices or returns, especially in the literature about stop-loss rules, are of the form of a random walk or a pure ARMA model (see, for example Kaminski and Lo [2014] and Acar and Toffel [2001]). Seeking more sophisticated models that may potentially better reflect the nature of equity prices, we present two models: the first one without a trend component (but with positive drift), and the second one allowing the possibility of a (temporary) trend component in the form of an ARMA. The modeling process consists of two parts conceptually: (1) modeling the standard deviation of returns, including white noise; (2) modeling the actual return.

Our analysis is based on the 30 NYSE stocks that were most traded in 2013. For these simulations, we consider hourly quotes from March 13 to March 24 2017 (both included), for each of the 30 stocks, which equates to 2070 hourly returns.

3.1 Modeling standard deviation and white noise

We modeled the standard deviation of the hourly returns with a GARCH(1,1) model. This requires that we model the white noise component as well. The GARCH(1, 1) model for the standard deviation σ_t, at each time t, is defined by the following equation

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2\sigma_{t-1}^2 + \beta\sigma_{t-1}^2$$

where ϵ_t is the random shock or innovation, occurring at time t, and which itself follows a certain conditional distribution that we must specify and calibrate with the observed data.

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In order to fit the best GARCH(1,1) possible, we try each of the implemented conditional distributions in the package **fGarch** [Wuertz et al., 2016] in R, fit the model, record its corrected Akaike Information Criterion (AIC), and eventually choose the conditional distribution (for the white noise) that provides the best (i.e. lowest) AIC on average over our 30 selected stocks. We find that the best conditional distribution is the Generalized Error Distribution (GED) with or without skewness parameter, depending whether we consider Model 1 or Model 2 (see next subsections).

A point worth advancing is that both in Model 1 and Model 2 the shape coefficient of the above conditional distribution is around 1.45, which is almost exactly half way between that of a Laplace distribution (shape=1) and a Gaussian distribution (shape=2), the former having a much larger kurtosis than the latter.

3.2 Model 1: Generalized Normally Distributed returns

Our first model for the hourly price of a hypothetical NYSE stock considers that the corresponding returns follow a Generalized Normal distribution. Recall that the PDF of the Generalized Normal (or Error) distribution is given by

$$GED(\mu, \sigma, \beta) = \frac{\beta}{2\sigma\Gamma(1/\beta)} e^{-(|x-\mu|/\sigma)^{\beta}}$$

where μ, σ, β are the location, scale, and shape parameters respectively, and Γ denotes the Gamma function. As Nadarajah [2005] showed, the PDF of a Generalized Normal distribution is just the same as the PDF of a Normal distribution but with a general β coefficient that is not necessarily equal to 2 (and hence the absolute value function is needed), as well as a different regularization term premultiplying the exponential function, so that the PDF integrates to unity.

After a model selection procedure to fit the parameters of the distributions involved, we obtain the following model to simulate the hourly price of our hypothetical NYSE stock.

$$P_{t-1} * (1 + r_t)$$

where

$$\begin{aligned} r_t &\sim GED(3.943 * 10^{-5}, 0.0042, 1.326), \\ \sigma_t^2 &\sim GARCH(1,1) \text{ with } \epsilon_t \sim SGED(\mu = 0, \sigma = 1, 1.4, 0.928) \end{aligned} \tag{6}$$

3.3 Model 2: ARMA returns

Our second model for the price of a hypothetical stock that trades in the NYSE considers that the returns follow an ARMA(p, q) process, with GARCH volatility. After the model selection procedure we obtain the following model for the hourly price of our stock:

$$P_{t-1} * (1 + r_t)$$

where

$$r_t \sim ARMA(3,0),\ \sigma_t^2 \sim GARCH(1,1) \text{ with } \epsilon_t \sim GED(\mu = 0, \sigma = 1, 1.47) \tag{7}$$

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4 Models of asset prices with overnight gap

Because we are introducing overnight gaps, the modeling process consists now of three parts: (1) modeling the overnight gap; (2) modeling the standard deviation of returns, including white noise; (3) modeling the actual return.

Part (2) of the modeling process will be the same as for the models without overnight gaps, since we want to consider the same modeling for the noise, and assess the impact of considering overnight gaps. Thus, we only describe the modeling of overnight gaps and their inclusion in the model for the returns.

4.1 Modeling overnight gaps

An overnight gap is defined as the difference between the Open price at period t, and the Close price at period $t-1$. Using our empirical data on 204,120 gaps (27 years worth of data on 30 NYSE stocks), we find that the average gap is about 1.000463 (i.e. +0.0463%). Furthermore, on average, gaps of any size above a minimum threshold of 0.1% happen around 77.93% of the days, and this is the frequency we will use in our simulations.

Including overnight gaps into the analysis of stop-loss rules is of critical importance, for two main reasons. First, overnight gaps can severely affect the profitability of an investing strategy that uses stop-loss orders, because they render the stop-loss barrier completely unable to protect the investor from a loss that is substantially larger than expected. In particular, this implies that by disregarding the phenomenon of overnight gaps one might inflate the usefulness of a stop-loss rule, and this is something ex-ante we wish to avoid. Second, overnight gaps are a very frequent phenomenon in the financial markets, and their magnitude can be non-negligible. For example, Figure 1 shows several overnight gaps in the candlestick plot of the price history of the American International Group (AIG:NYSE) from 28th of August to 22nd of September, 2017. In this plot on 28th Sept. the Close price is 60.66 and next session Open price (on 29th Sept.) is 60.20, a fall of 0.7%; on 8th Sept. (a Friday) the Close is 59.78 and next session on 11th Sept. the Open is 60.77, a jump of 1.6%. All in all, by considering overnight gaps we obtain a more reliable assessment of stop-loss performance.

We modeled overnight gaps as a Weibull random variable for the following reasons. We think of overnight gaps as extreme price movements, and the Weibull is one of the three families of extreme value distributions. Moreover, a convenient way to reproduce the impact of overnight gaps in the price of a stock is to see the gap as a scaling factor of the price, and so we need a distribution with non-negative support, as it is the Weibull, because the scaled stock price must naturally remain non-negative. Finally, the Weibull distribution is very flexible, capable of showing a completely different density plot depending on the value of its two parameters: scale (λ) and shape (κ).

Recall that the probability density function (PDF) of a Weibull random variable is

$$Weibull(x;\lambda,\kappa) = \begin{cases} \frac{\kappa}{\lambda}\left(\frac{x}{\lambda}\right)^{\kappa-1} e^{-(x/\lambda)^{\kappa}}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $\lambda > 0$ is the scale parameter and $\kappa > 0$ is the shape parameter of the distribution. When the random variable is clear from context it is customary to write $Weibull(\lambda,\kappa)$. We follow a 3-step procedure to optimize the Weibull parameters:

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Figure 1: AIG candlestick plot of price from 28/08 to 22/09 showing overnight gaps (source: Yahoo Finance).

1. Explore seven estimation methods to find a first approximation: Method of Moments, Maximum Likelihood Estimation, Median and Quartiles, Mean and Standard Deviation, Mean Rank, Median Rank, and Symmetric CDF.

2. Select the best estimation method and do a grid search around the estimated parameters to improve *Stress*. *Stress* is a measure of goodness of fit that we define as a convex combination of the D statistic from the Kolmogorov-Smirnov test and the Mean Squared Error, in proportion 3:7 as the latter metric is more robust.

3. Correction for Survivorship Bias: the 30 stocks we have picked are the ones that have "survived" from 1990 to 2017, and so it is advisable to be more conservative in asserting the efficacy of stop-loss rules. This is achieved by increasing the downside risk of the overnight gap, which in the case of a Weibull random variable this may be achieved by reducing the shape parameter of this distribution. We perform this step with care: we require that the new calibration delivers an average gap that is more negative than the average gap observed empirically, but with the least increase in Stress possible with respect to Step 2.

As it can be observed in Table 1, the value of the optimized parameters according to the three steps above are $\lambda = 1.0033$, and $\kappa = 170.7193$, delivering a Stress level of 0.0363.

4.2 Model 1: Generalized Normally Distributed returns with gap

In this subsection we introduce overnight gaps into the first parametric model presented earlier.

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Table 1: Weibull parameter calibration

Method	Step 1 λ, κ, Stress	Step 2 λ, κ, Stress	Step 3 λ, κ, Stress
MoM	1.0063, 98.56, 0.0677	–	–
MLE	1.1743, 20.16, 0.4790	–	–
MQ	1.0018, 203.62, **0.0385**	1.0028, 189.32, 0.0244	1.0033, 170.72, 0.0363
MSD	1.0055, 112.53, 0.0586	–	–
MR	1.0075, 81.35, 0.0812	–	–
MDR	1.0075, 81.47, 0.0811	–	–
SCDF	1.0075, 81.49, 0.0811	–	–

Again, after a model selection procedure to fit the parameters of the distributions involved, we obtain the following model to simulate the hourly price of our hypothetical NYSE stock.

$$P_t = \begin{cases} P_{t-1} * Gap_t, & \text{for } t \bmod 7 = 1 \\ P_{t-1} * (1 + r_t), & \text{otherwise} \end{cases} \tag{9}$$

where the Gap_t occurs daily (t mod 7 = 1) with a certain probability (again, in our case, 77.93%)

$$Gap_t = \begin{cases} 1, & \text{if no gap occurs} \\ Weibull(1.0033, 170.7193), & \text{if a gap occurs} \end{cases} \tag{10}$$

and

$$\begin{aligned} r_t &\sim GED(3.943 * 10^{-5}, 0.0042, 1.326), \\ \sigma_t^2 &\sim GARCH(1,1) \text{ with } \epsilon_t \sim SGED(\mu = 0, \sigma = 1, 1.4, 0.928) \end{aligned} \tag{11}$$

As it can be seen, the value of the parameters is the same compared to the model without gap. This is consistent for two reasons. First, because the Weibull distribution is approximately centered at unity. Second, because the impact of overnight gaps on return volatility is carefully taken into account at a later stage (see Section 4.4).

4.3 Model 2: ARMA returns with gap

In this case we are introducing the overnight gap into our second parametric model.

After the model selection procedure we obtain the following model for the hourly price of our stock:

$$P_t = \begin{cases} P_{t-1} * Gap_t, & \text{for } t \bmod 7 = 1 \\ P_{t-1} * (1 + r_t), & \text{otherwise} \end{cases} \tag{12}$$

where Gap_t is defined as in Equation 10, and

$$r_t \sim ARMA(3,0),\ \sigma_t^2 \sim GARCH(1,1) \text{ with } \epsilon_t \sim GED(\mu = 0, \sigma = 1, 1.47) \tag{13}$$

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4.4 Incorporating post-gap effect on volatility

In addition, and both for Model 1 and Model 2, we consider that a stock's volatility at time t is a function of the magnitude of the most recent gap, and so we attempt to model this relationship in order to obtain an even more realistic model.

With this in mind, we fit a polynomial in the gap magnitude to the *relative range* (a proxy for volatility to be made precise below). We observed that terms of order higher than three of the variable "gap magnitude", although significantly different from 0 in a statistical sense were practically insignificant in magnitude (in the order of 10^{-6}). For this reason, we propose the following model for the impact of the gap magnitude on the relative range:

$$RelRange = \beta_0 + \beta_1 gm + \beta_2 gm^2 + \beta_3 gm^3 + u \tag{14}$$

where gm is the gap magnitude, and the relative range of the stock on a given day, and the relative range, referred to as $RelRange$, is defined as $\frac{Range}{Low} = \frac{High - Low}{Low}$.

Next, we compute the average relative range for our stocks and compute the relative betas, which are the original betas divided by the average relative range. Then, the relative betas are averaged across stocks. From here, we obtain two standard deviation multipliers:

$$NGM = \overline{rel.beta_0}$$

and

$$GM = \overline{rel.beta_0} + \overline{rel.beta_1} * gm + \overline{rel.beta_2} * gm^2 + \overline{rel.beta_3} * gm^3$$

where the horizontal bars signal that the *average* relative beta has been considered, while NGM and GM are the standard deviation multipliers when there is no gap and when there is, respectively. Those quantities squared multiply the GARCH-based variance that Models 1 and 2 produce at a given time t, displayed in Equation 11 and Equation 13 respectively. In particular, we find that, in expectation

$$NGM = 0.80 \tag{15}$$

and

$$GM = 0.80 + 0.25gm + 0.06gm^2 - 0.01gm^3 \tag{16}$$

Considering that the price gap is a real number generally close to unity, we find that a higher price gap in magnitude is associated with higher volatility, as measured by the relative range.

5 Non parametric approach: the stationary bootstrap.

There is no one universally accepted model for asset returns, and whatever this model might be the danger of overfitting is always present. In light of those reasons, we complement our analysis with a non parametric approach to add robustness: the stationary bootstrap. The stationary bootstrap [Politis and Romano, 1994] is one of the existing block bootstrap methods, which is characterized by using random block lengths that are distributed according to the geometric distribution. Provided that the original data is stationary, the use of geometrically distributed block lengths ensures that the resampled series remain stationary. By using an appropriate scenario generator for

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financial time series such as the stationary bootstrap, our backtesting is more reliable than if we just used the observed prices, as these are just one possible realization of an underlying stochastic process.

The stationary bootstrap algorithm goes as follows (cf. [Davison and Hinkley, 1997, Ch. 8]): Given $X_1, \ldots, X_n$ observations of a time series, wrap this data around a circle; that is, define $Y_i = X_{1+(i-1 \mod n)}$ and $Y_0 = X_n$. Let $I_1, I_2, \ldots \in \{1, \ldots, n\}$ be drawn iid with uniform distribution (these are the starting points of blocks). Construct a bootstrap sample as follows:

- set $Y_1^* = Y_{I_1}$.
- For $i = 2, \ldots, n$, let $Y_i^* = Y_{I_i}$, with probability p, and let $Y_i^* = Y_{j+1}$ with probability $1 - p$, where $Y_{i-1}^* = Y_j$.

The output is a series $Y_1^*, \ldots, Y_n^*$, composed of blocks with mean length $1/p$, since the lengths $L_1, L_2, \ldots$ of attached blocks through the iteration follows a geometric distribution: $\mathbb{P}(L = k) = p(1-p)^{k-1}$, $k = 1, 2, \ldots$.

In our stationary bootstrap sampling of time series data we make use of the automatic block-length selection heuristic laid out in [Politis and White, 2004]. Also, to be rigorous with the application of the sampling method, we checked for weakly stationarity in the observed data, applying traditional tests for unit root (Augmented Dickey Fuller, KPSS, Phillips-Perron) and a more advanced test, namely, the Priestley-Subba Rao (PSR) test for nonstationarity [Priestley and Rao, 1969], which is based upon examining how homogeneous a set of spectral density function (SDF) estimates are across time, across frequency, or both. Thus, we determine that at the 1% significance level, eleven stocks (BAC, PFE, F, C, JPM, MRK, XOM, AIG, HAL, KEY, and BMY) present weakly stationary hourly returns in the period considered: March 13 to May 26, 2017.

Finally, we note that under this nonparametric approach, our price model is

$$P_t = P_{t-1}(1 + r_t^*)$$

for all $t \geq 1$, where r_t^* is the bootstrapped return at time t. Figure 2 shows real price time series (continuous, red line) and bootstrapped versions (dashed, black line) for Citigroup and American Int. Group.

6 Model-based simulation

We present in this section our experimental results for the different model-based simulations of the four stop-loss rules considered: the fixed percentage, ATR, RSI and triple MA crossover rules. First we specify the choice of parameters for each of the stop-loss rules.

6.1 Parameter choice for the fixed percentage rule

As seen in Section 2, we implement the following stop-loss:

$$SL_t = P_{t-1}^{max}(1 - a)$$

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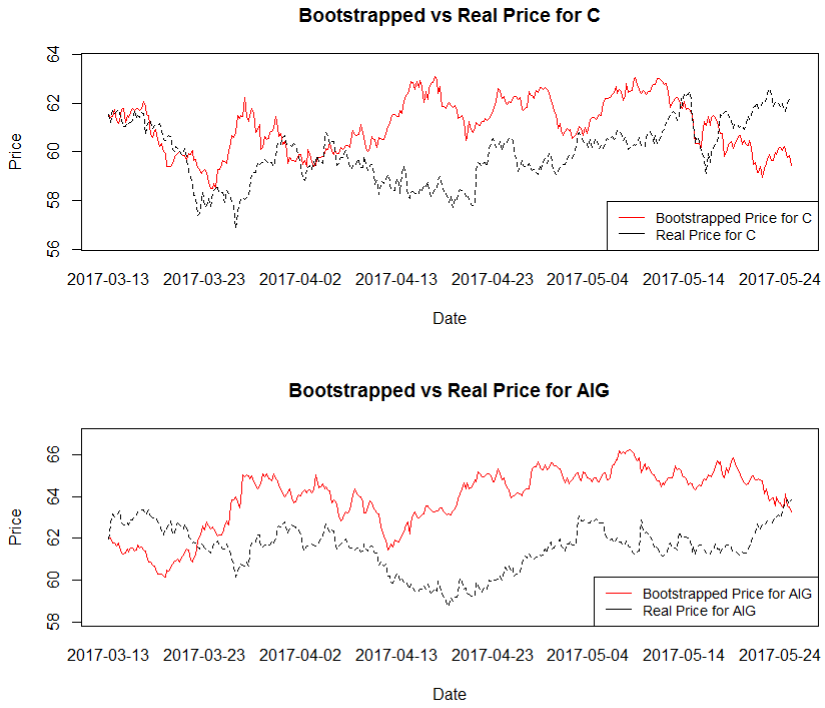


Figure 2: Citigroup (C) and AIG bootstrapped price and real price time series.

where $a > 0$ is the maximum percentage of the highest price achieved until time $t-1$ that the investor is willing to lose in that operation.

There seems not to be a consensus among practitioners regarding the most suitable choice of value for the parameter of this stop-loss rule. Hence, instead of arbitrarily choosing the parameter's value that determines how tight the exit barrier is placed around a reference price, we decided to tune the parameter to a value that delivers a good out-of-sample stop-loss performance. In order to accomplish this, we have a range in the interval $[0.03, 0.1]$ and select the value that improves the performance of the rule the most on average, in terms of expected return, Sharpe Ratio, Sortino Ratio, Return-VaR ratio, and Return-ES ratio.

6.2 Parameter choice for the ATR, RSI and triple MA crossover rules

On the other hand, there seems to be a consensus among practitioners regarding suitable parameter values for these three other stop-loss rules, and for that reason we have chosen not to tune those parameters and instead use some of the standard values used in practice. We then consider:

- ATR

$$SL_t = P_{t-1} - 2.5ATR_t$$

With a time frame of 14 days, as is standard practice.

- RSI

$$SL_t := RSI_t(7) \geq 70$$

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As shown above, we decided to use the standard value of 70 as a threshold signaling that the stock has become overbought and a price reversal may follow soon. The chosen time frame of 7 days is also common among practitioners.

- Triple MA crossover

$$SL_t := MA(5) < MA(20) \wedge MA(20) < MA(70)$$

We use standard time frames of 5, 20, and 70 trading sessions (days). These figures are, however, modified in our non-parametric model to accommodate for the shorter period considered (due to constraints on high-frequency data availability), and they become instead 3, 12, and 30 trading days.

6.3 Simulation results

Next we present the results of our Monte Carlo simulations, based on 10,000 repetitions in each case. Because we repeat each set of simulations 50 times, we are able to use the Welch's t-test to assess the statistical significance of the difference in mean values that the stop-loss rules provide compared to Buy-and-Hold. We use the following notation to indicate statistical significance:
ns: p-value ≥ 0.05, *: $0.01 \leq$ p-value < 0.05, **: $0.001 \leq$ p-value < 0.01, ***: p-value < 0.001

Tables 2, 3, 4, and 5 show the out-of-sample results for two of the four stop-loss rules, comparing stop-loss performance both without and with overnight gaps. Results for the other two rules are presented in the Appendix (Tables 9, 10, 11 and 12)

Table 2: Fixed percentage SL, without gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	RVaR	RES
SL - Model 1	4.57%	0.81	0.24	0.85	0.75
B&H - Model 1	7.09%	0.32	0.34	0.16	0.14
Difference	-2.52% ns	+0.49***	-0.10***	+0.69***	+0.61***
SL - Model 2	4.51%	0.82	0.24	0.86	0.76
B&H - Model 2	6.82%	0.34	0.33	0.18	0.15
Difference	-2.31% ns	+0.48***	-0.09***	+0.68***	+0.61***

Table 3: Fixed percentage SL, with gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	RVaR	RES
SL - Model 1	4.04%	0.81	0.14	0.84	0.69
B&H - Model 1	4.99%	0.29	0.17	0.15	0.12
Difference	-0.95% ns	+0.52***	-0.03***	+0.69***	+0.57***
SL - Model 2	4.06%	0.81	0.14	0.83	0.69
B&H - Model 2	4.93%	0.30	0.18	0.16	0.13
Difference	-0.87% ns	+0.51***	-0.04***	+0.67***	+0.56***

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Table 4: RSI-based SL, without gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	R-VaR	R-ES
SL - Model 1	6.27%	0.33	0.27	0.22	0.19
B&H - Model 1	7.04%	0.32	0.32	0.16	0.13
Difference	-0.77% ns	+0.01 ns	-0.05***	+0.06***	+0.06***
SL - Model 2	6.47%	0.35	0.30	0.22	0.18
B&H - Model 2	6.87%	0.34	0.33	0.18	0.15
Difference	-0.40% ns	+0.01**	-0.03***	+0.04**	+0.03***

Table 5: RSI-based SL, with gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	R-VaR	R-ES
SL - Model 1	4.51%	0.31	0.14	0.24	0.20
B&H - Model 1	4.99%	0.29	0.18	0.15	0.12
Difference	-0.48% ns	+0.02***	-0.04***	+0.09***	+0.08***
SL - Model 2	4.57%	0.32	0.15	0.23	0.19
B&H - Model 2	4.94%	0.30	0.18	0.16	0.13
Difference	-0.37% ns	+0.02***	-0.03***	+0.07***	+0.06***

As a general conclusion, both in the models without and with gap, we observe that under both Model 1 and 2, there exist stop-loss rules that provide a higher risk-adjusted return than Buy-and-Hold (B& H). The results are particularly remarkable under Model 1: Sharpe Ratio almost triplicates, while RVaR and RES are roughly multiplied by five.

Note that in a rising market (with expected annual return of around 5% in our models including overnight gaps), stop-loss rules seem to provide a slightly lower expected return in absolute terms, but this difference is not statistically significant and in any case, it is on average small (usually less than 100 bps).

Further note that the usefulness of each stop-loss rule is similar in both models, being the fixed percentage stop-loss the generally most useful strategy.

Nonetheless, it is worth pointing out that stop-loss performance – with perhaps the exception of the fixed percentage rule – is in general somewhat worse under the no-gap models. This fact may seem counterintuitive at first because one may reasonably think that since large negative overnight gaps are able to bypass stop-loss rules, rendering the investor unable to liquidate his/her position at the desired price level, a model that does not include this kind of phenomena should inflate stop-loss performance. However, this reasoning relies on the inaccurate assumption that large negative overnight gaps are sufficiently frequent. In fact, large negative overnight gaps are much less common than what it might appear; it is just that because when they occur they attract considerable attention, individuals tend to overestimate their frequency or probability (*availability heuristic*). On the other hand, overnight gaps small in magnitude (negative or positive) are relatively frequent, and it seems that some of the stop-loss rules considered are able to avoid the negative ones while benefiting from the positive ones, improving

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risk-adjusted performance.

By not including overnight gaps when modeling stock returns, it seems that one is not getting the most out of stop-loss rules. This, together with our use of high-frequency data, more complex price models and alternative risk-adjusted performance metrics, might explain why the literature on stop-loss rules, which at present does not consider overnight gaps, has not found a significant performance improvement in the use of stop-loss rules, despite the fact that in practice they are considered an extremely effective tool for loss protection, in particular, and risk management, in general.

Therefore, since real financial markets quite frequently exhibit overnight gaps, and assuming that our overnight gap specification is sufficiently accurate, we believe that the results we obtained with the models that do include gaps are closer to the true stop-loss performance than those obtained under the no-gap models.

For the ATR and MA based stop-loss rules, we obtained again statistically significant risk-adjusted performance improvements, albeit more moderate. See Tables 9, 10, 11, and 12.

7 Data-based simulation results

Table 6: Bootstrap simulation results under Buy-and-Hold

Stock	$E[r]$	Sharpe	Sortino	RVaR	RES
BAC	−7.1794	−0.6747	−0.6273	−0.4822	−0.3943
PFE	−5.6818	−1.4347	−0.8502	−0.8827	−0.7053
F	−12.8653	−1.6387	−0.8669	−1.0756	−0.8700
C	1.5171	0.1578	0.1423	0.0741	0.0603
JPM	−6.1557	−0.8472	−0.7039	−0.5708	−0.4628
MRK	−0.3745	−0.0722	−0.2421	−0.0623	−0.0502
XOM	0.1034	0.0182	0.1392	−0.0061	−0.0049
AIG	3.3076	0.4296	0.6963	0.2527	0.2053
HAL	−9.5564	−0.8444	−0.6948	−0.6144	−0.5040
KEY	−2.8110	−0.2334	−0.3376	−0.1981	−0.1626
BMY	−7.1723	−1.1828	−0.7980	−0.7790	−0.6288

First of all, it must be stressed that in the period March 13 to May 26, all the stocks under consideration, except Citigroup (C) and American International Group (AIG), experienced a negative return, which explains why most of the expected returns are negative, especially under Buy-and-Hold. Because the metrics besides expected return that we have considered are meaningless when below zero, we restrict our attention to expected return for those stocks. In such a situation in which 9 out of the 11 stocks have experienced a negative return in the period considered, the stop-loss rules have provided a higher expected return, in some cases extremely higher than if no stop-loss had been used, as in the case of the fixed percentage strategy, which has achieved a positive expected return in 5 out of 11 stocks, and near positive in two more stocks. In fact, on average, the fixed percentage rule has increased the expected return in over 4 percentage points compared to Buy-and-Hold, and in some cases (Ford), as much as

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Table 7: Bootstrap simulation results under fixed % SL

Stock	$E[r]$	Sharpe	Sortino	RVaR	RES
BAC	−0.6170	−0.1950	−0.3535	−0.7837	−0.6448
PFE	−1.9672	−1.4759	−0.8676	−2.3369	−1.8209
F	−1.6234	−0.7815	−0.7160	−1.3576	−0.9843
C	1.2653	0.1405	0.2264	−0.0147	−0.0142
JPM	−0.9784	−0.3912	−0.5309	−1.1617	−0.9611
MRK	−0.0666	−0.0212	−0.1208	−0.5641	−0.4970
XOM	0.1272	0.0373	0.0153	−0.4631	−0.4304
AIG	3.1447	0.4063	0.8429	0.2378	0.2248
HAL	−0.7096	−0.2295	−0.3914	−0.8943	−0.7409
KEY	−0.1206	−0.0322	−0.1312	−0.6302	−0.5238
BMY	−1.3588	−0.6840	−0.6881	−1.6237	−1.4127

Table 8: Bootstrap simulation results under RSI SL

Stock	SL.expret	SL.Sharpe	SL.Sortino	SL.RVAR	SL.RES
BAC	−6.9053	−0.6416	−0.5863	−0.4686	−0.3827
PFE	−5.6759	−1.4283	−0.8299	−0.8873	−0.7088
F	−12.6731	−1.5550	−0.8480	−1.0665	−0.8647
C	1.4081	0.1656	0.2199	0.0899	0.0735
JPM	−5.9739	−0.8039	−0.6615	−0.5584	−0.4526
MRK	−0.3390	−0.0654	−0.1249	−0.0524	−0.0422
XOM	0.1185	0.0226	0.0096	0.0082	0.0066
AIG	3.0064	0.4306	0.8039	0.2746	0.2244
HAL	−9.2652	−0.8016	−0.6611	−0.6053	−0.4965
KEY	−2.6207	−0.2209	−0.2805	−0.1815	−0.1491
BMY	−7.1070	−1.1565	−0.7738	−0.7727	−0.6253

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11.3 percentage points higher than Buy-and-Hold, on average (in 200,000 repetitions), in *just* two and a half months. In a full year, the improvement might be even more spectacular. This is the opposite case as in our model-based simulation: in the present case, using real (bootstrapped) prices, because mid March to late May has been a bad period for most of the stocks under consideration, it seems indeed natural that a strategy that at some point switches to a risk-free asset will perform better than merely Buy-and-Hold. However, three aspects must be stressed: first, the risk-free rate used in the bootstrap simulations is the 3-month U.S. Treasury bill rate that was in place in March 2017, just 0.74%; second, the fact that the remarkable results shown in Table 7 for the falling stocks are, again, obtained out of sample; lastly, in the case of the fixed percentage rule, note that the expected return premium over Buy-and-Hold in a Bear market is much higher than the expected return premium of Buy-and-Hold over this stop-loss rule in a Bull market, in general.

On the other hand, for C and AIG, since most of the metrics are above 0, we can conduct a similar analysis as in Section 6. In this case the results are favorable too, albeit less spectacular: certain stop-loss rules seem to provide a modest improvement in risk-adjusted returns, at the expense of a very slight drop in absolute expected return. For example, as shown in Table 8 we may highlight the improvement in *all* risk-adjusted metrics that the RSI-based stop-loss provided for the two rising stocks (C and AIG).

The simulation results for the other two stop-loss rules (ATR and MA) can be found in the Appendix at the end of this work (Tables 13 and 14).

8 Conclusions

Combining two different parametric price models, together with a nonparametric approach based on the stationary bootstrap as a scenario generating method, our out-of-sample research shows that irrespective of the market situation (a rising or falling market), some stop-loss rules are able to provide a higher expected return than buy-and-hold, at least when adjusting for risk, and this difference is statistically significant at the 0.1% level in most of the cases. On the one hand, in the case of a rising market, certain stop-loss rules provide a better risk-adjusted return than Buy-and-Hold under most risk-adjusted metrics. On the other hand, in the case of a falling market, all stop-loss rules provide a better expected return than Buy-and-Hold. In the case of the fixed percentage rule, the improvement in expected return in a falling market is outstanding in many cases, and the improvement in risk-adjusted return in a rising market is often remarkable as well.

These results show that by incorporating a relatively simple risk management mechanism such as stop-loss rules it is possible sometimes to significantly enhance risk-adjusted investment performance, introducing very few changes to an existing buy-and-hold passive strategy.

We also showed that by not including overnight gaps, stop-loss performance might in fact be reported to be lower than it truly is, which might be one of the factors that explains the general skepticism towards this risk management mechanism present in the academic literature, despite its undisputed popularity among practitioners.

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Appendix

Below are the model-based simulation results for stop-loss rules ATR (Tables 9 and 10) and MA (Tables 11 and 12), followed by the data-based (i.e. bootstrapped) simulation results for stop-loss rules ATR (Table 13) and MA (Table 14).

Table 9: ATR-based SL, without gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	R-VaR	R-ES
SL - Model 1	6.49%	0.32	0.31	0.15	0.13
B&H - Model 1	7.23%	0.32	0.34	0.16	0.14
Difference	-0.74% ns	0.00 ns	-0.03**	-0.01*	-0.01**
SL - Model 2	5.59%	0.33	0.26	0.16	0.13
B&H - Model 2	6.84%	0.33	0.33	0.17	0.14
Difference	-1.25% ns	0.00 ns	-0.07***	-0.01**	-0.01***

Table 10: ATR-based SL, with gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	R-VaR	R-ES
SL - Model 1	4.03%	0.33	0.12	0.16	0.13
B&H - Model 1	4.99%	0.29	0.18	0.15	0.12
Difference	-0.96% ns	+0.04***	-0.06***	+0.01***	+0.01*
SL - Model 2	3.98%	0.35	0.12	0.18	0.14
B&H - Model 2	4.96%	0.30	0.18	0.16	0.13
Difference	-0.98% ns	+0.05***	-0.06***	+0.02***	+0.01***

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Table 11: MA crossover-based SL, without gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	RVaR	RES
SL - Model 1	5.73%	0.32	0.27	0.16	0.12
B&H - Model 1	7.12%	0.32	0.33	0.16	0.13
Difference	-1.39% ns	0.00 ns	-0.06***	0.00 ns	-0.01**
SL - Model 2	5.49%	0.34	0.26	0.18	0.14
B&H - Model 2	6.79%	0.33	0.33	0.18	0.15
Difference	-1.30% ns	+0.01 ns	-0.07***	0.00 ns	-0.01***

Table 12: MA crossover-based SL, with gap: Simulation results

Strategy-Model	$\mathbb{E}$(r)	Sharpe	Sortino	RVaR	RES
SL - Model 1	4.28%	0.31	0.13	0.16	0.13
B&H - Model 1	4.91%	0.28	0.17	0.15	0.12
Difference	-0.63% ns	+0.03***	-0.04***	+0.01***	+0.01***
SL - Model 2	4.30%	0.32	0.14	0.17	0.13
B&H - Model 2	4.95%	0.30	0.18	0.16	0.12
Difference	-0.65% ns	+0.02***	-0.04***	+0.01***	+0.01*

Table 13: Bootstrap simulation results under ATR SL

Stocks	SL.expret	SL.Sharpe	SL.Sortino	SL.RVAR	SL.RES
BAC	−6.5103	−0.6435	−0.5897	−0.4400	−0.3545
PFE	−4.9619	−1.3276	−0.8114	−0.7354	−0.5841
F	−10.6830	−1.4408	−0.8287	−0.8277	−0.6647
C	1.4632	0.1553	0.2321	0.0682	0.0554
JPM	−5.7171	−0.8118	−0.6655	−0.5255	−0.4231
MRK	−0.3448	−0.0679	−0.1313	−0.0702	−0.0560
XOM	0.1091	0.0206	0.0123	−0.0067	−0.0054
AIG	3.1674	0.4244	0.8606	0.2512	0.2036
HAL	−9.2054	−0.8241	−0.6708	−0.5857	−0.4796
KEY	−2.5792	−0.2233	−0.2888	−0.1909	−0.1549
BMY	−6.8442	−1.1392	−0.7689	−0.7225	−0.5839

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Table 14: Bootstrap simulation results under MA SL

Boot.Perf.df	SL.expret	SL.Sharpe	SL.Sortino	SL.RVAR	SL.RES
BAC	−6.8231	−0.6512	−0.5913	−0.4218	−0.3451
PFE	−5.4746	−1.3589	−0.8166	−0.8202	−0.6593
F	−12.4149	−1.5483	−0.8457	−1.0095	−0.8203
C	1.4312	0.1535	0.2289	0.0652	0.0526
JPM	−5.8740	−0.8127	−0.6639	−0.5030	−0.4099
MRK	−0.3471	−0.0678	−0.1297	−0.0540	−0.0432
XOM	0.1154	0.0220	0.0107	−0.0043	−0.0034
AIG	3.1591	0.4191	0.8656	0.2377	0.1904
HAL	−9.1302	−0.8119	−0.6641	−0.5451	−0.4500
KEY	−2.6658	−0.2268	−0.2914	−0.1760	−0.1439
BMY	−6.8560	−1.1209	−0.7631	−0.7010	−0.5701

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